

Econ 131
Spring 2026
Danny Yagan

Problem Set 1

DUE: 02-25-2026 5PM PT on Gradescope

Student Name: *Marcus Huang*

Student ID: 3041432436

- Write or type your answers clearly and in dark ink (physical or electronic ink) so that your responses are legible.
- Tag each of your answers on Gradescope so that it is clear what responses are to which questions.
- **Although you may work in groups**, each student must submit individual sets of solutions. You must note the names of other students that you worked with. Write their names here:

1. True/False Statements [30 points]

Determine whether each statement is true, false, or uncertain and explain why. Answers with no explanation will receive no points. [5 points each]

- (a) Difference-in-differences yields the causal effect of interest if the treatment group and the control group have identical outcome levels before the treatment begins.

- I think this statement is *False*.
- Difference-in-differences does not require the treatment and control groups to have the same starting outcome levels. Instead, it relies on the parallel trends assumption, which states that in the absence of the treatment, the outcomes for both groups would have trended similarly over time.
- A difference-in-differences analysis is valid even there is a pre-existing level difference, as long as that difference remains constant except for the policy change.

- (b) In January 2025, the Congressional Budget Office released its updated budget outlook. Download it [here](#). Look at the top half of Table B-1. Compute the average interest rate on government debt in 2035 r_{2035} by dividing 2035 Net Interest by 2034 Debt Held By The Public. Compute the 2035 GDP growth rate g_{2035} by comparing 2034 GDP to 2035 GDP. True/False/Uncertain: If the interest rate on government debt (r_t) and the GDP growth rate (g_t) stay constant forever at their 2035 levels, the government can run a permanent primary deficit while maintaining a stable debt to GDP level.

• I think this statement is True.

• $r_{2035} : \frac{\text{2035 net interest}}{\text{2034 debt held by the public}} \approx 3.6\%$

• $g_{2035} : \frac{\text{GDP}_{2035} - \text{GDP}_{2034}}{\text{GDP}_{2034}} \approx 3.8\%$

• Based on the CBO projections, $r(3.6\%) < g(3.8\%)$ in 2035. Under these constant conditions, the government can run a permanent primary deficit while maintaining a stable debt to GDP level.

(c) Adult income differences across U.S. local areas and across U.S. racial and ethnic groups are explained by differences in how rich or poor their parents were.

- Intuitively False !

- I think the parental income level alone can not explain these gaps between racial and ethnic groups and geography difference.

- These gaps might related to specific events or significance more .

For areas , different states have its own laws and taxes . Some parental income level also may show different living quality vary from states .

(d) Suppose that a government applies a 10% income tax and uses the proceeds to give everyone a \$10,000 cash transfer. This tax-and-transfer system is progressive.

- I think this is True.

- In this scenario, a flat tax with a universal transfer is mathematically progressive because the constant transfer makes up a much larger percentage of total income for the poor than for the rich.

(e) If a government places a \$1 per unit tax on producers of cigarettes rather than on consumers, producers will bear a larger economic burden of the tax than they would if the tax were placed on consumers instead.

- This is false I think.

- The economic incidence is independent of the statutory incidence.

The burden is determined by market elasticities, meaning the outcome is the same whether the tax is placed on who.

(f) When a sales tax is made more visible and salient to consumers at the point of purchase, it breaks the standard tax incidence result that equilibrium is independent of whether the tax is nominally placed on consumers or producers.

- This statement is True.
- Just like the example on lecture, when the price with tax is visible to the consumer, they act purchase less.
- This behavioral response breaks the standard theory that the market equilibrium is unaffected by the nominal placement of tax.

2. Difference-in-Differences (DiD) and Parallel Trends [30 points]

A policy is implemented in the **treated** group starting in **2015**. The **control** group is not directly affected. Let $Y_{g,t}$ denote the average outcome for group $g \in \{\text{Treated}, \text{Control}\}$ in year t .

Throughout, use:

Pre period = 2010–2014, Post period = 2015–2020.

The DiD estimand is:

$$\hat{\tau}_{DiD} = \left(\bar{Y}_{T, \text{Post}} - \bar{Y}_{T, \text{Pre}} \right) - \left(\bar{Y}_{C, \text{Post}} - \bar{Y}_{C, \text{Pre}} \right).$$

Scenario 1 [18 points]

Table 1: Scenario 1 outcomes, 2010–2020

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Control ($Y_{C,t}$)	50	51	52	53	54	55	56	57	58	59	60
Treated ($Y_{T,t}$)	55	56	57	58	59	62	63	64	65	66	67

- (a) [4 points] Compute the four group means: $\bar{Y}_{T, \text{Pre}}$, $\bar{Y}_{T, \text{Post}}$, $\bar{Y}_{C, \text{Pre}}$, $\bar{Y}_{C, \text{Post}}$. Show your work.

$$\cdot \bar{Y}_{T, \text{Pre}} = (55 + 56 + 57 + 58 + 59) \div 5 = 285 \div 5 = 57$$

$$\cdot \bar{Y}_{T, \text{Post}} = (62 + 63 + 64 + 65 + 66 + 67) \div 6 = 387 \div 6 = 64.5$$

$$\cdot \bar{Y}_{C, \text{pre}} = (50 + 51 + 52 + 53 + 54) \div 5 = 260 \div 5 = 52$$

$$\cdot \bar{Y}_{C, \text{post}} = (55 + 56 + 57 + 58 + 59 + 60) \div 6 = 345 \div 6 = 57.5$$

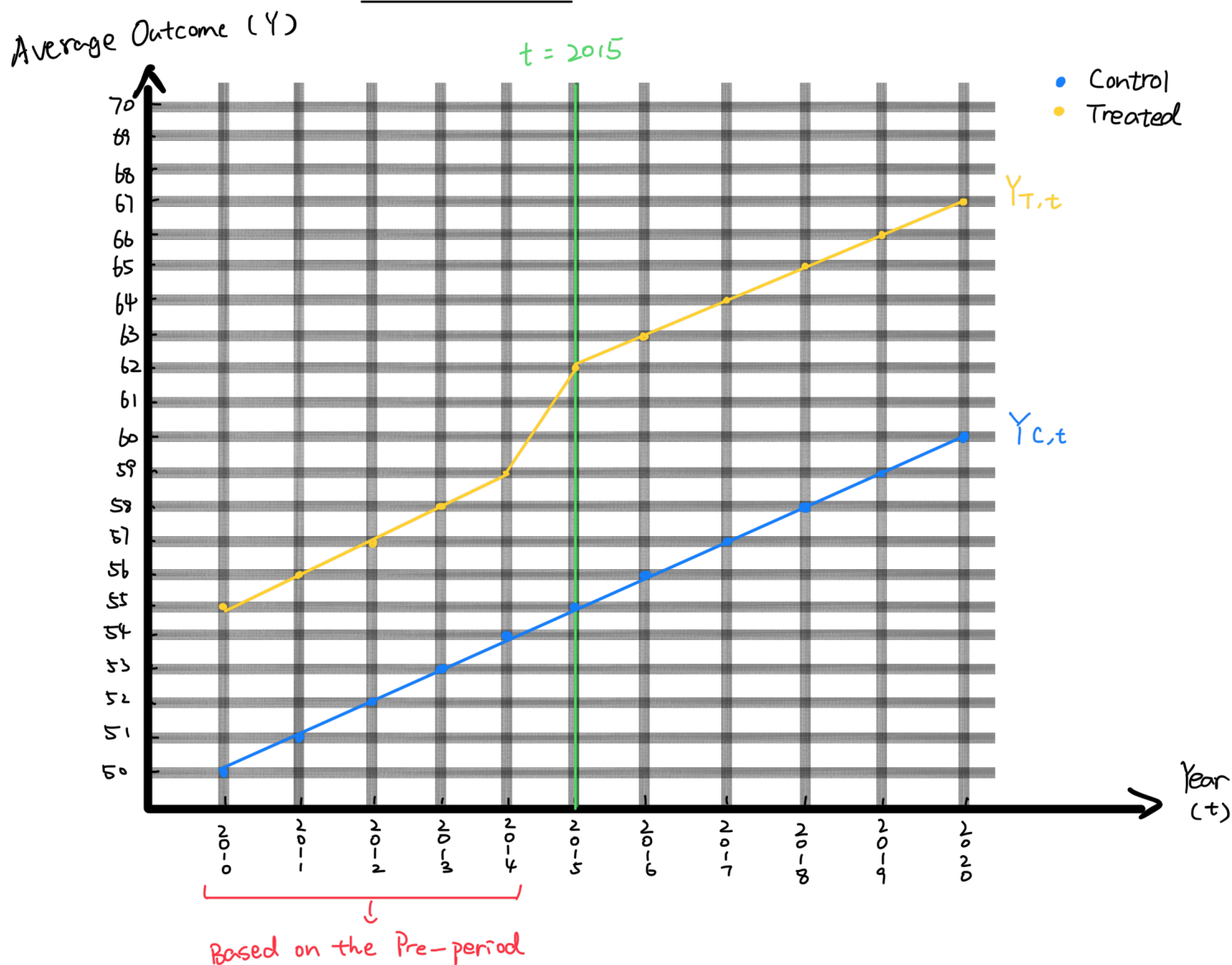
(b) [4 points] Compute the changes within each group: $\Delta_T = \bar{Y}_{T,Post} - \bar{Y}_{T,Pre}$ and $\Delta_C = \bar{Y}_{C,Post} - \bar{Y}_{C,Pre}$.

- Treated Change : $\Delta_T = 64.5 - 57 = 7.5$
- Control Change : $\Delta_C = 57.5 - 52 = 5.5$

(c) [5 points] Compute $\hat{\tau}_{DiD} = \Delta_T - \Delta_C$. In 1-2 sentences, explain what this number means.

- $\hat{\tau}_{DiD} = \Delta_T - \Delta_C = 7.5 - 5.5 = 2$
- The DiD estimand is 2. This number represents the estimated causal effect of the policy. It implies that the policy caused the outcome for the treated group to increase by 2 units more than it would have without the intervention.

- (d) [5 points] On the same axes, plot $Y_{T,t}$ and $Y_{C,t}$ for $t = 2010, \dots, 2020$. Draw a vertical line at 2015. Based only on 2010–2014, does parallel trends look plausible?



- Yes, parallel trends look plausible. Both groups were following identical increment 1 unit per year before the policy began in 2015.

Scenario 2 [12 points]

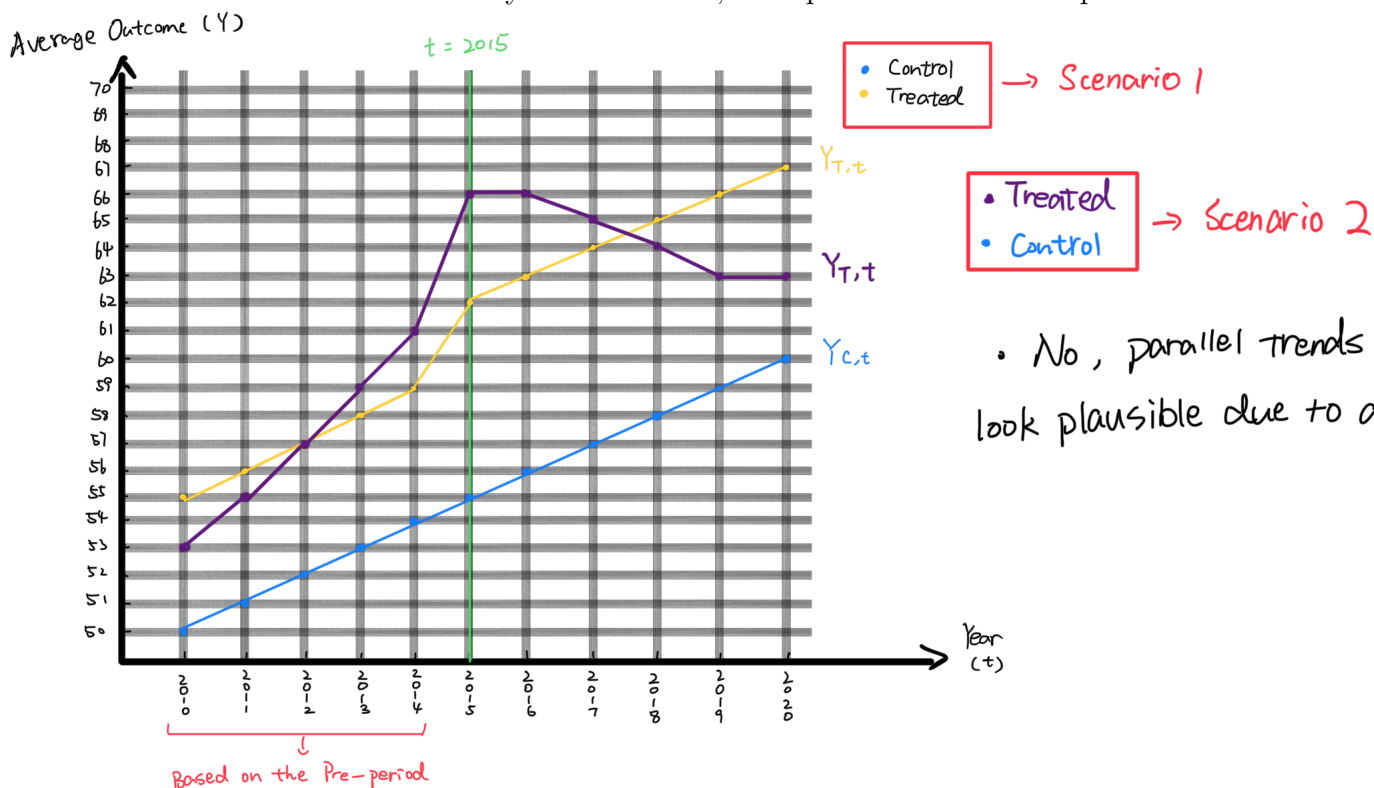
Table 2: Scenario 2 outcomes, 2010–2020 (same DiD as Scenario 1, but violates parallel trends)

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Control ($Y_{C,t}$)	50	51	52	53	54	55	56	57	58	59	60
Treated ($Y_{T,t}$)	53	55	57	59	61	66	66	65	64	63	63

(e) [5 points] Repeat the same steps as Scenario 1. Compute $\bar{Y}_{T,Pre}$, $\bar{Y}_{T,Post}$, $\bar{Y}_{C,Pre}$, $\bar{Y}_{C,Post}$, then Δ_T , Δ_C , and finally $\hat{\tau}_{DiD}$.

- $\bar{Y}_{T,Pre} = (53 + 55 + 57 + 59 + 61) \div 5 = 285 \div 5 = 57$
- $\bar{Y}_{T,Post} = (66 + 66 + 65 + 64 + 63 + 63) \div 6 = 387 \div 6 = 64.5$
- $\bar{Y}_{C,Pre} = 52$, $\bar{Y}_{C,Post} = 57.5$
- $\Delta_C = 57.5 - 52 = 5.5$, $\Delta_T = 64.5 - 57 = 7.5$
- DiD Estimand = $7.5 - 5.5 = 2$

- (f) [4 points] On the same axes, plot $Y_{T,t}$ and $Y_{C,t}$ for $t = 2010, \dots, 2020$. Draw a vertical line at 2015. Based only on 2010–2014, does parallel trends look plausible in Scenario 2?



- No, parallel trends do not look plausible due to different slopes.

- (g) [3 points] Even if the average DiD number matches Scenario 1, why is Scenario 2 more concerning? (1–2 sentences.)

- Scenario 2 is concerning because it violates the identifying assumption of parallel trends. We cannot distinguish the causal effect of the policy from pre-existing differences in the groups' growth trends.

3. Tariffs and Welfare [28 points]

Figure 1 displays the case of a small open economy facing a world price and imposing a tariff.

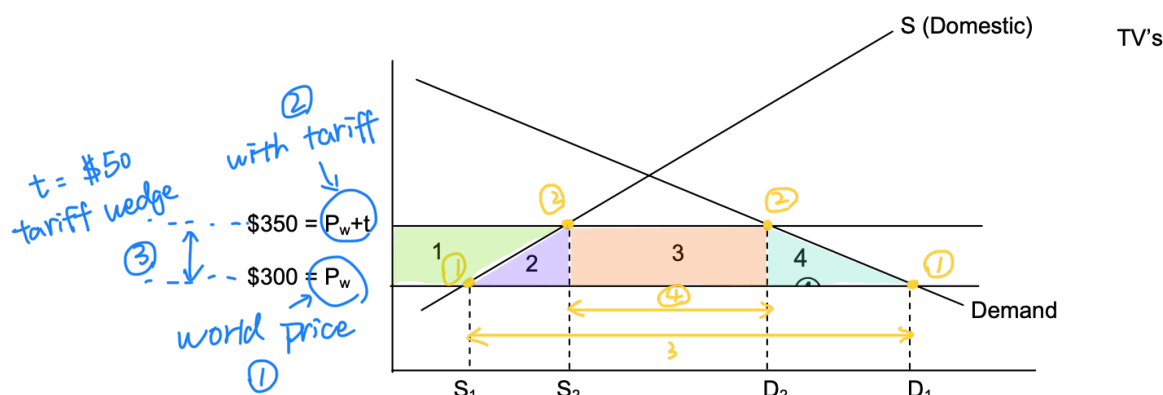


Figure 1: Small Country Case

- (a) [2 points] On the diagram, circle and label the world price pre-tariff, the tariff-inclusive domestic price, and draw a vertical bracket showing the size of the tariff wedge.

• In the diagram above.

- (b) [3 points] Label the domestic quantity supplied and domestic quantity demanded both before and after the tariff. On the horizontal axis, mark the import quantity before the tariff and after the tariff.

• In the diagram above.

① D_1 and S_1

② D_2 and S_2

③ $D_1 - S_1$

④ $D_2 - S_2$

- (c) [8 points] The lecture diagram labels four areas (1, 2, 3, 4). For each numbered area: shade/outline the area clearly, write what it represents (e.g. “producer surplus gain”), state who gets it (consumers, producers, government), and say whether it is a gain or loss for the home country. [2 points per area]

- Area 1 : producer surplus gain , the domestic producers get it .

It is a gain for the home country .

- Area 2 : production distortion , no one get this , like DWL .

It is a loss for the home country .

- Area 3 : Government Revenue , the government gets it .

It is a gain for the home country .

- Area 4 : Consumption Distortion , no one get this , like DWL .

It is a loss for the home country .

(d) [3 points] Identify which areas are the deadweight loss triangles in the small-country case.

Next to each triangle, write a short phrase explaining the economic distortion it captures: one triangle from too much domestic production (production distortion), and one triangle from too little domestic consumption (consumption distortion).

- Area 2 is the production distortion. It captures the economic waste from domestic overproduction. The country is spending more money to produce goods at home than it would cost to import them.

- Area 4 is the consumption distortion. It captures the economic waste from domestic under-consumption. Consumers stop buying units that they value more than the true world price, resulting in a loss of total welfare.

- (e) [8 points] Now consider the following specific example. A small country has domestic supply and demand curves for steel (quantities in thousands of tons):

$$Q^D = 100 - P$$

$$Q^S = P$$

where P is the price in dollars per ton. The world price of steel is $P_W = \$20$ per ton.

- i. [2 points] Without a tariff, compute the domestic quantity demanded, domestic quantity supplied, and imports. Show your work.

- Domestic quantity demanded = $Q^D = 100 - P_W = 100 - 20 = 80$
- Domestic quantity supplied = $Q^S = P_W = 20$
- Imports = $Q^D - Q^S = 80 - 20 = 60$

- ii. [2 points] Now suppose the government imposes a specific tariff of $t = \$10$ per ton on imported steel. Compute the new domestic price, domestic quantity demanded, domestic quantity supplied, and imports. Show your work.

- New domestic price = $P_w + t = 20 + 10 = \$30$
- Domestic quantity demanded = $Q^D = 100 - P = 100 - 30 = 70$
- Domestic quantity supplied = $Q^S = P = 30$
- New imports = $Q^D - Q^S = 70 - 30 = 40$

iii. [2 points] Calculate the following welfare effects (show your calculations): consumer surplus loss, producer surplus gain, government tariff revenue, and total deadweight loss.

$$\bullet \text{ consumer surplus loss} = 0.5 \times (\text{old } D + \text{new } D) \times \text{tariff}$$

$$\Rightarrow 0.5 (80 + 70) \times 10 = 750$$

$$\bullet \text{ producer surplus gain} = 0.5 \times (\text{old } S + \text{new } S) \times \text{tariff}$$

$$\Rightarrow 0.5 (20 + 30) \times 10 = 250$$

$$\bullet \text{ government tariff revenue} = \text{tariff} \times \text{new imports}$$

$$\Rightarrow 10 \times 40 = 400$$

$$\bullet \text{ total deadweight loss} = \text{CSL} - \text{PSG} - \text{GR}$$

$$\Rightarrow 750 - 250 - 400 = 100$$

iv. [2 points] Verify that your deadweight loss equals the sum of the production distortion triangle and the consumption distortion triangle.

- Production distortion = $0.5 (\text{increase in supply}) \times \text{tariff}$

$$\Rightarrow 0.5 (30 - 20) \times 10 = 50$$

- Consumption distortion = $0.5 (\text{decrease in demand}) \times \text{tariff}$

$$\Rightarrow 0.5 (80 - 70) \times 10 = 50$$

- $DWL = PD + CD = 50 + 50 = 100$

$\Rightarrow$ correct.

- (f) [4 points] Now consider a large country that can affect world prices. In 3–4 sentences, explain how the analysis would differ. What additional welfare effect arises when a large country imposes a tariff, and why does this make the overall welfare impact ambiguous?

Unlike small countries, a large country's tariff reduces global demand enough to force foreign exports to lower their world prices P_w , creating a term-of-trade gain for the home country. This gain is a wealth transfer from foreigners to the domestic government, which can potentially offset the national inefficiency. However, the country still has DWLs, so the overall welfare impact is vague because the net result depends on whether the gain from foreign countries is larger than the domestic DWL.